

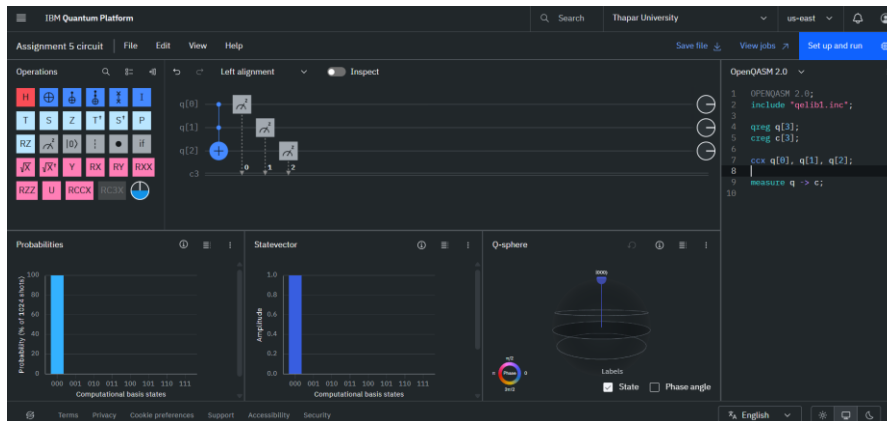
Assignment 5

Divjot Singh Gulati 102316025

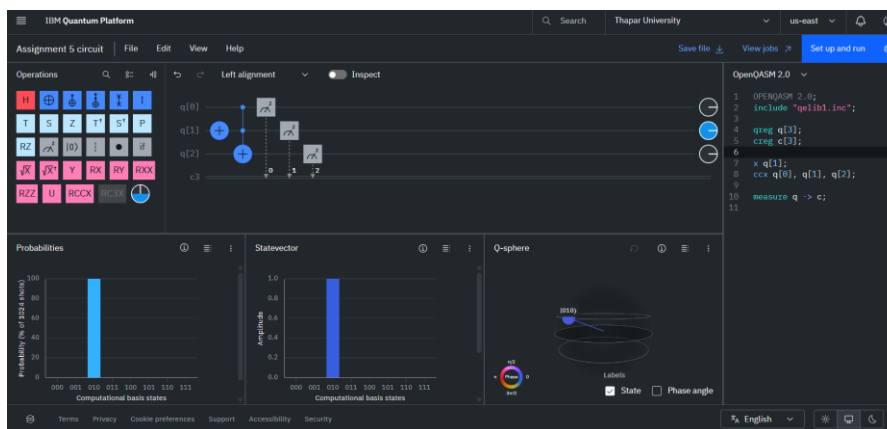
Experiment 1:-

Toffoli Gate as a Universal Classical Gate

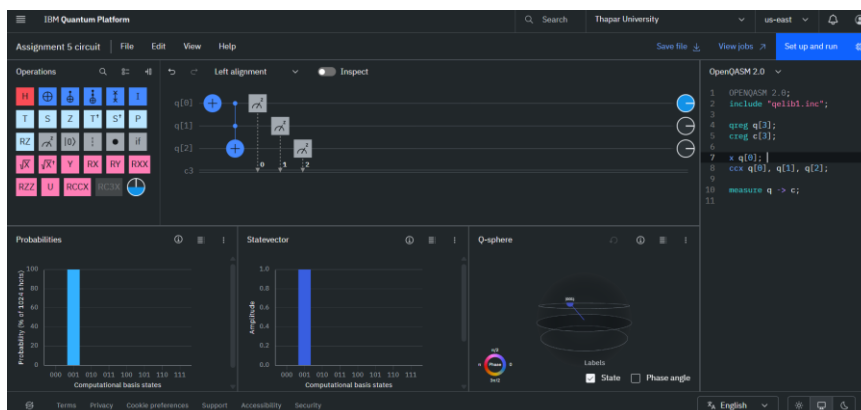
1. For AND Gate
 - a. When $A=0, B=0$



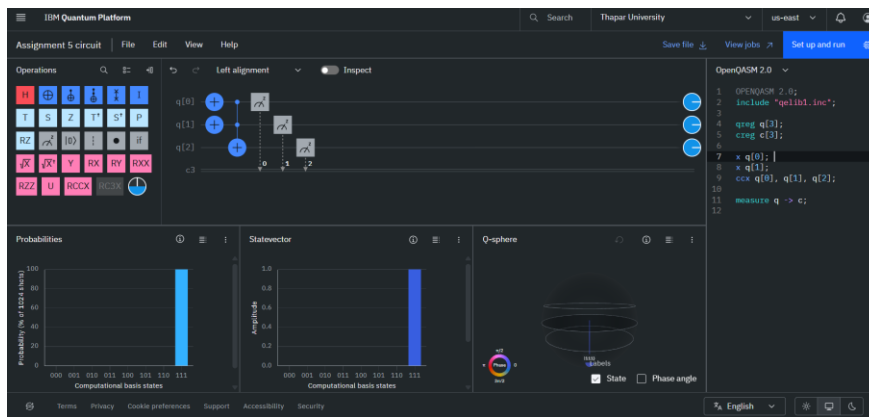
- b. When $A=0, B=1$



- c. When $A=1, B=0$

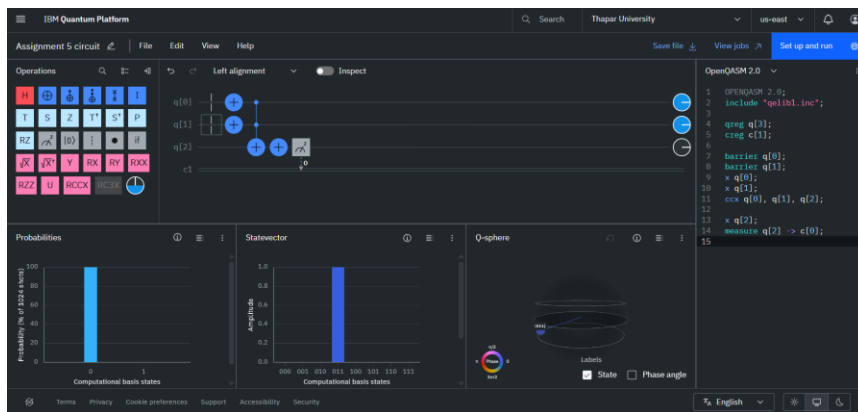


d. When $A=1, B=1$

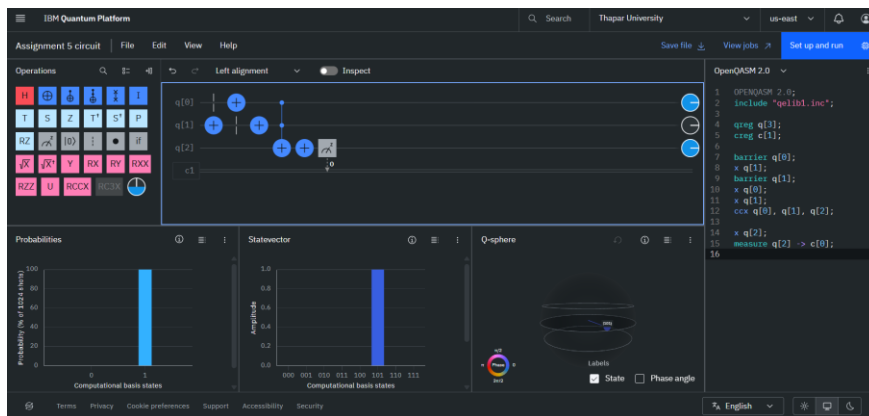


2. For OR Gate

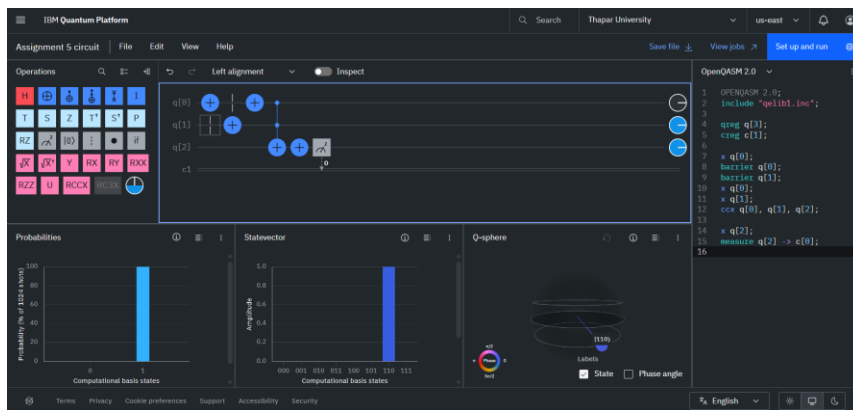
a. When $A=0, B=0$



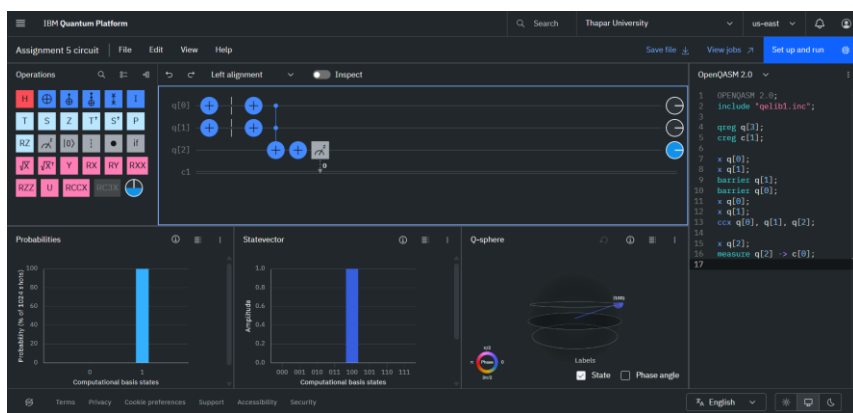
b. When $A=0, B=1$



c. When $A=1, B=0$

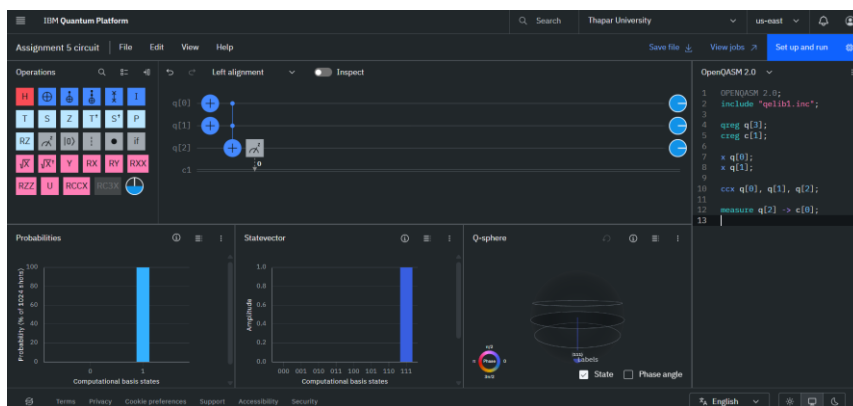


d. When $A=1, B=1$

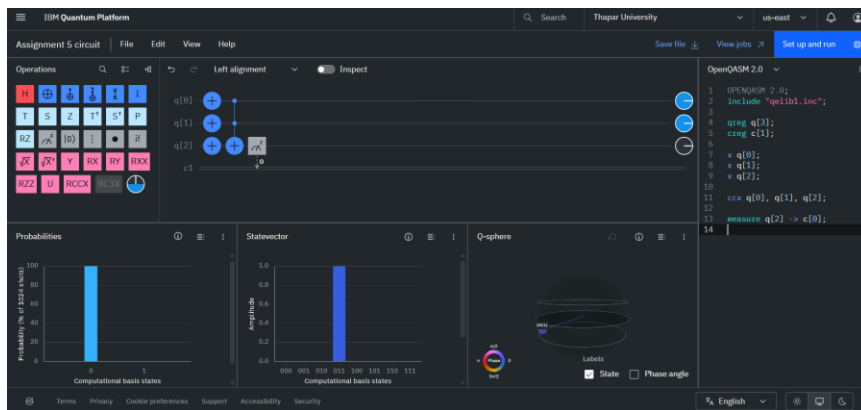


3. For NOT Gate

a. When $C=0$, we get output as 1



b. When $C=1$, we get output as 0

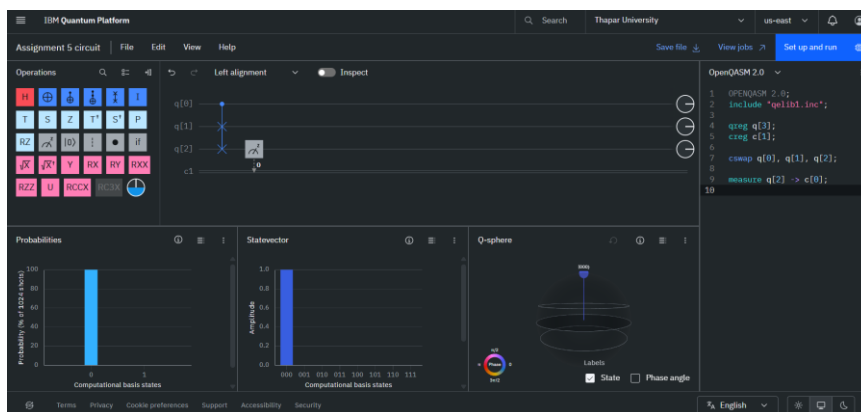


Experiment 2:-

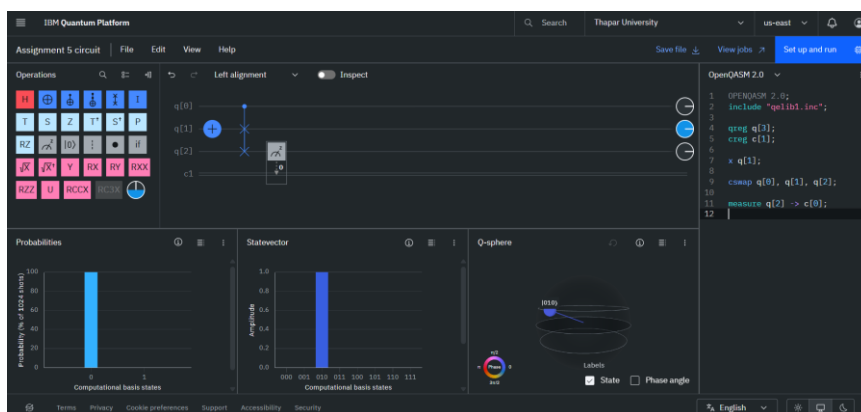
Fredkin Gate as a Universal Classical Gate

1. For AND Gate

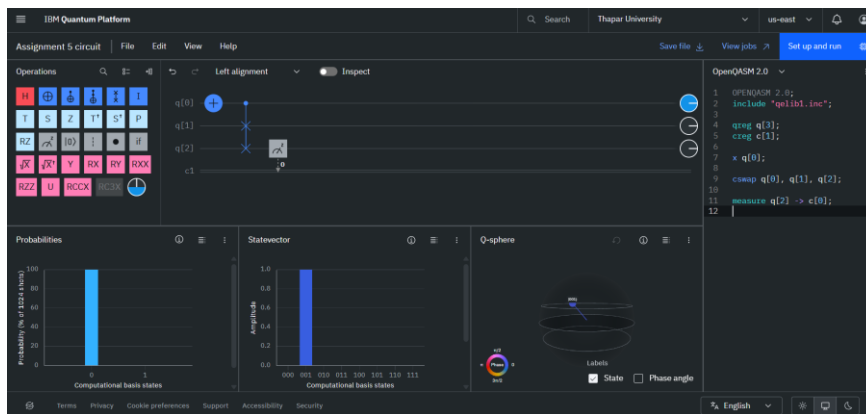
a. When $A=0$, $B=0$



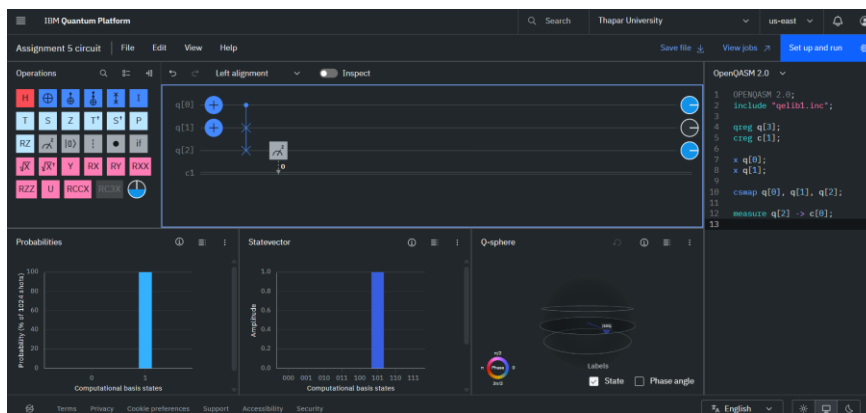
b. When $A=0$, $B=1$



c. When $A=1, B=0$

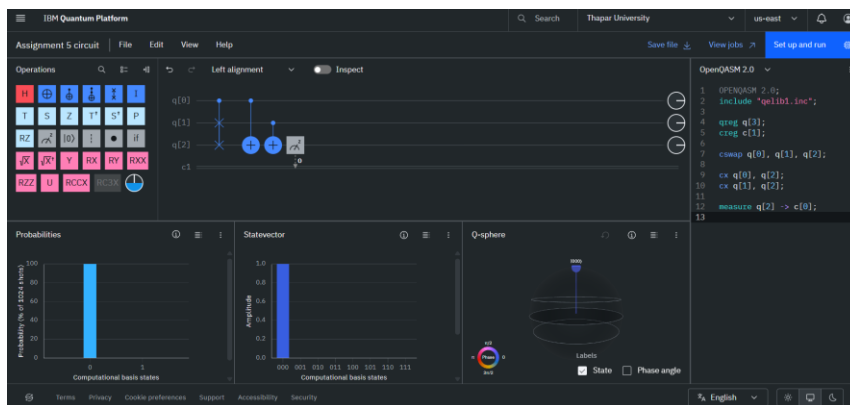


d. When $A=1, B=1$

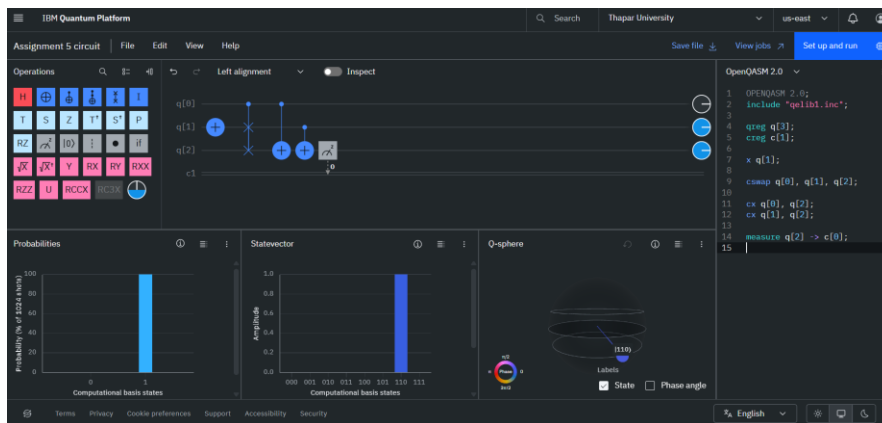


2. For OR Gate

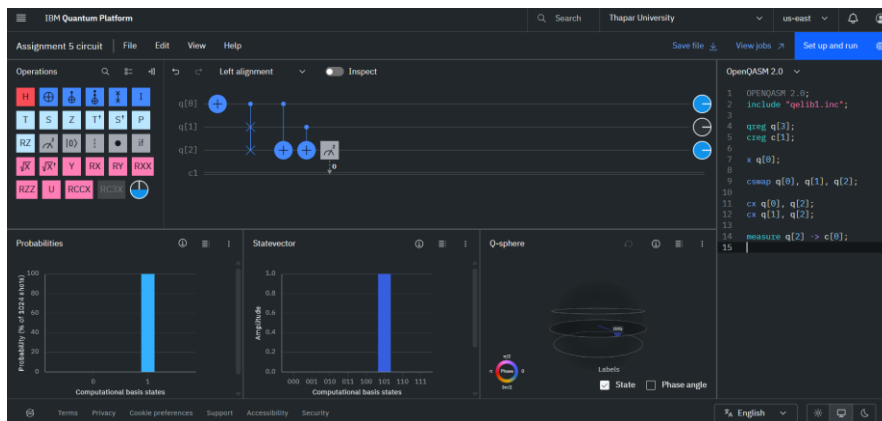
a. When $A=0, B=0$



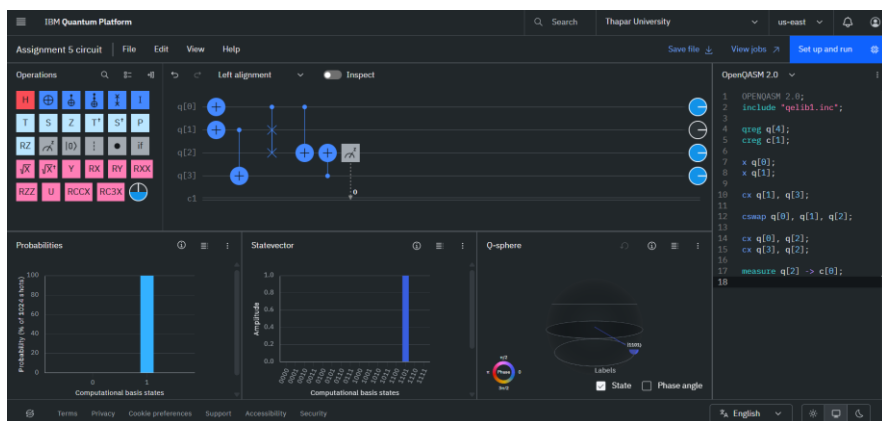
b. When $A=0$, $B=1$



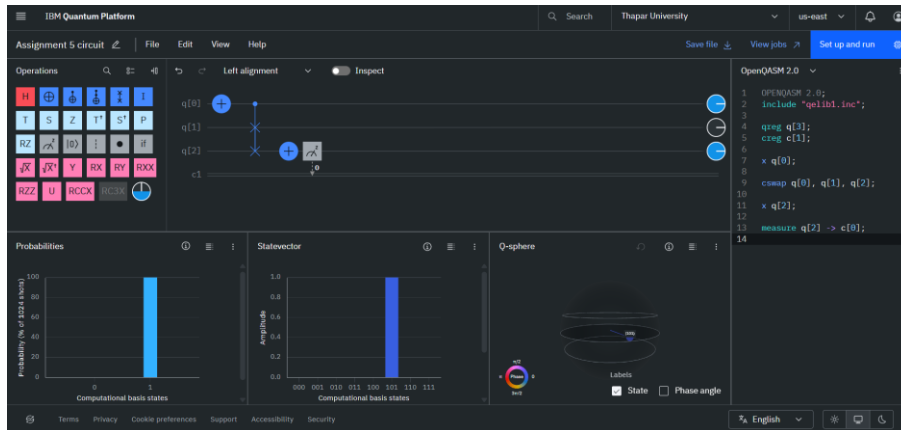
c. When $A=1$, $B=0$



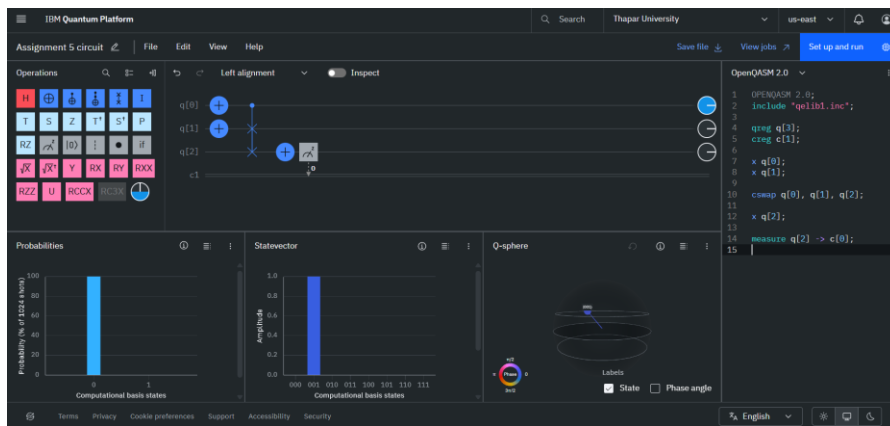
d. When $A=1$, $B=1$



3. For NOT Gate
 - a. When $C=0$, we get output as 1



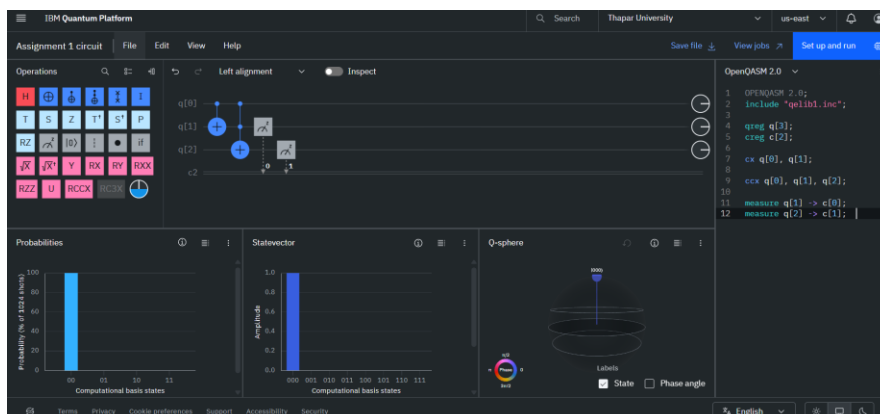
- a. When $C=1$, we get output as 0



Experiment 3:-

Half and Full Adder Quantum Circuits

1. Half Adder
 - a. When $A=0$, $B=0$

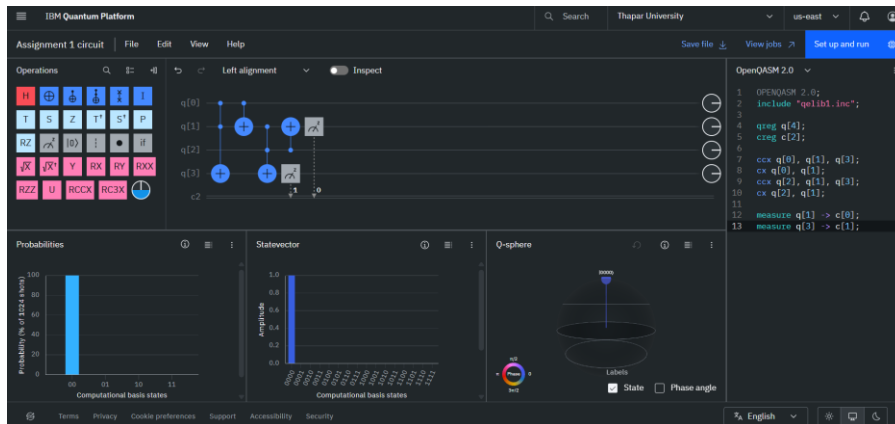


The screenshot displays the IBM Quantum Platform interface for a quantum circuit. The top bar shows the IBM Quantum Platform logo, a search bar, the user's name (Thapar University), and a 'Log out' button. Below the top bar, there are tabs for 'Assignment 1 circuit', 'File', 'Edit', 'View', and 'Help'. The main workspace is divided into several panels:

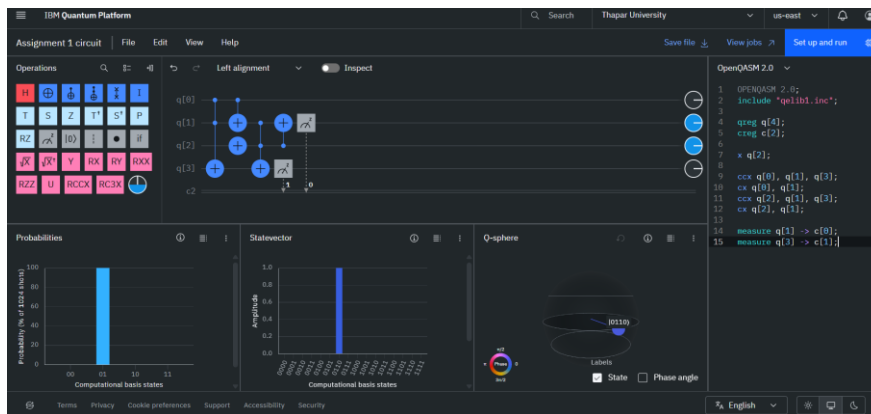
- Operations:** A panel on the left containing a grid of quantum gates (H, S, Z, T, P, RZ, X, Y, RY, RXX, RZZ, U, RXX, etc.) and a 'Left alignment' dropdown.
- Circuit Diagram:** A central area showing a quantum circuit with three qubits (q[0], q[1], q[2]). The circuit includes gates like H, S, Z, T, P, RZ, X, Y, RY, RXX, RZZ, U, RXX, and a 'Left alignment' dropdown.
- Probabilities:** A bar chart on the bottom left showing the probability of each computational basis state (000, 001, 010, 011, 100, 101, 110, 111). The state 100 has the highest probability, around 100%.
- Statevector:** A bar chart on the bottom middle showing the amplitude of each computational basis state. The state 100 has the highest amplitude, around 1.0.
- Q-sphere:** A Bloch sphere on the bottom right showing the state of the qubits. The state is labeled 'State' and 'Phase angle'.
- OpenQASM 2.0:** A code editor on the right showing the OpenQASM 2.0 code for the circuit.

2. Full Adder

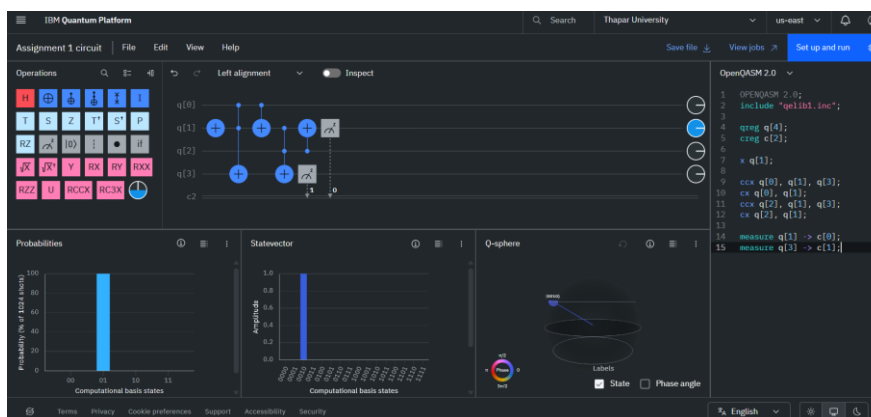
a. When $A=0, B=0$



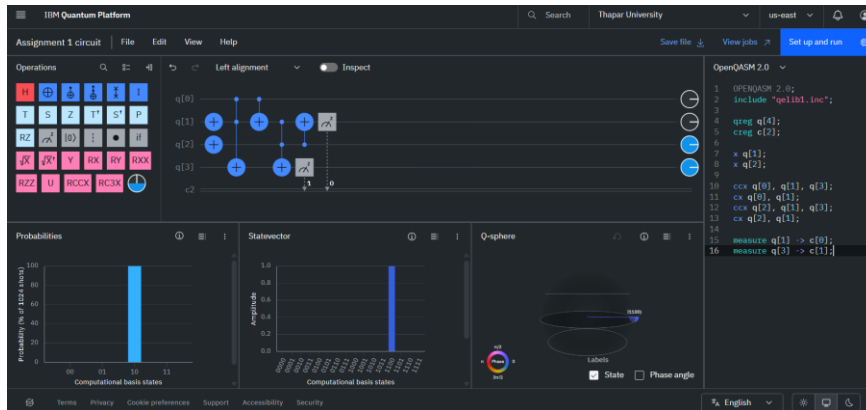
b. When $A=0, B=0, C_{in}=1$



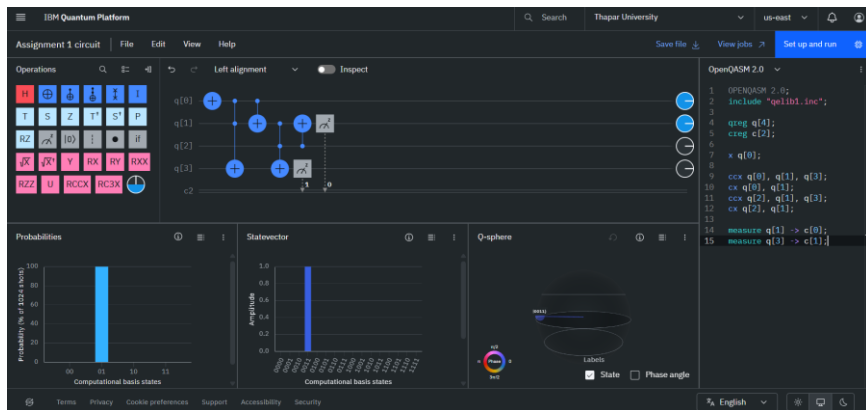
c. When $A=0, B=1, C_{in}=0$



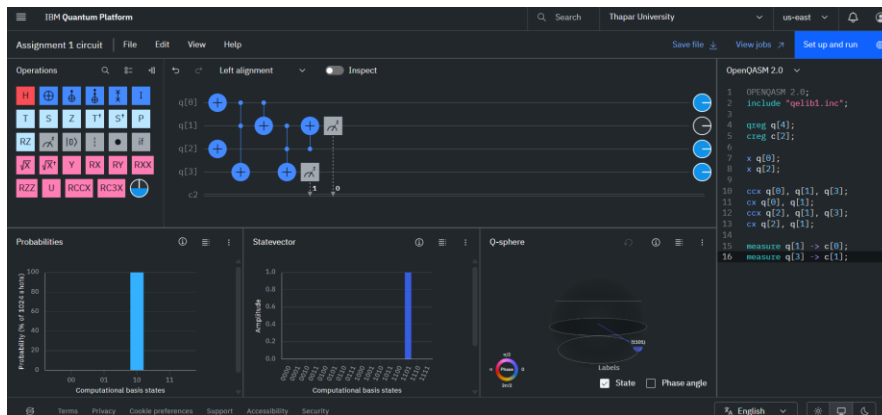
d. When $A=0$ $B=1$ $C_{in}=1$



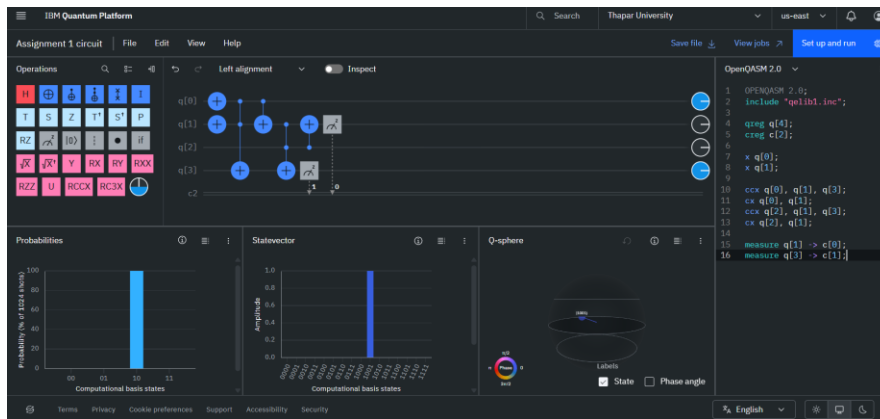
e. When $A=1$ $B=0$ $C_{in}=0$



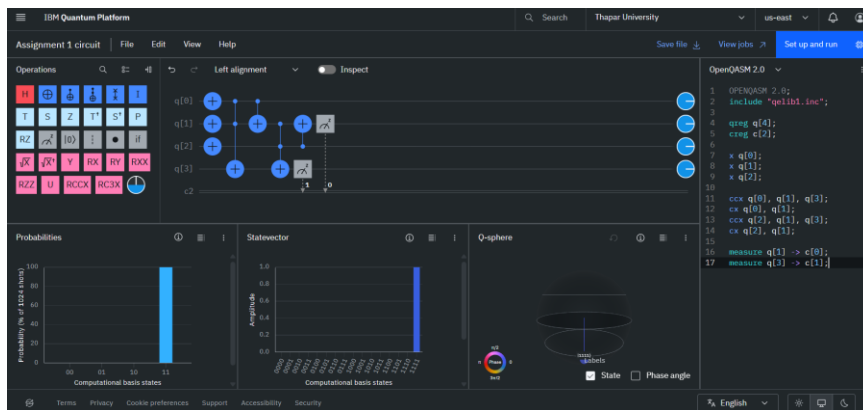
f. When $A=1$ $B=0$ $C_{in}=1$



g. When A=1 B=1 Cin=0



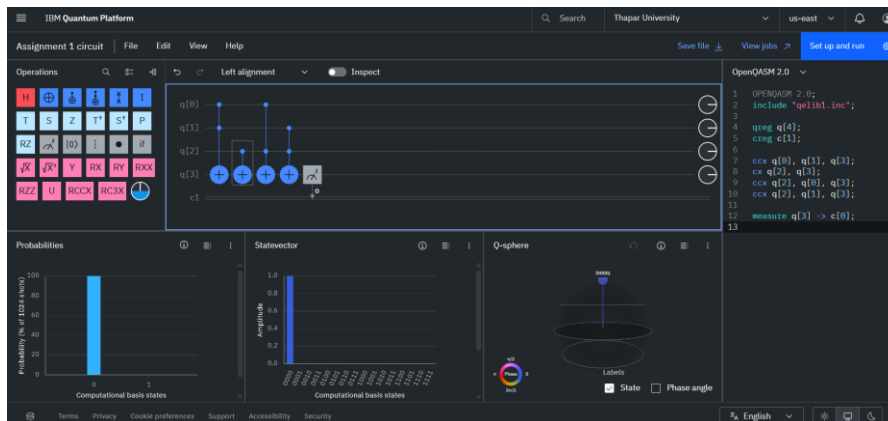
h. When A=1 B=1 Cin=1



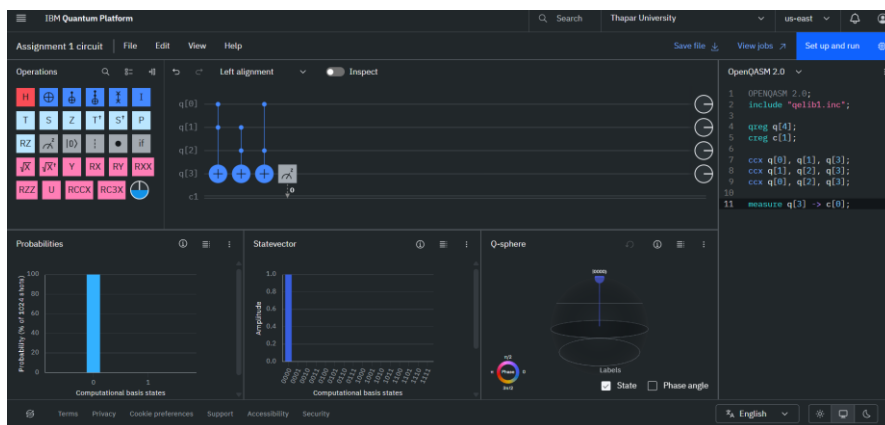
Experiment 4:-

Classical Boolean Function to Quantum Circuit Conversion

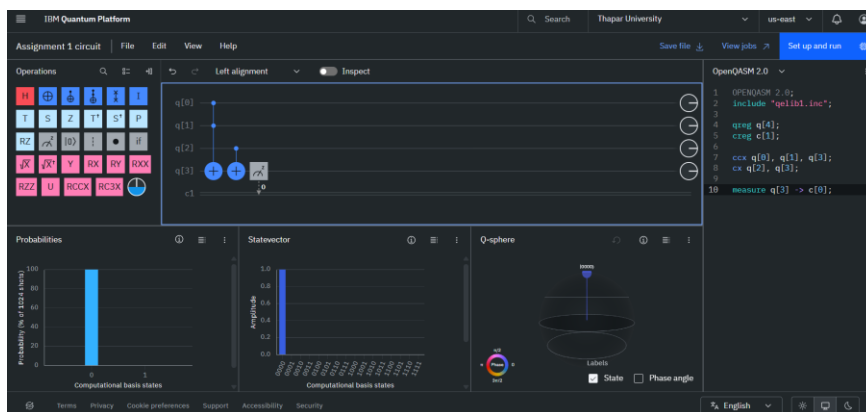
$$1. f(x,y,z) = (x \wedge y) \vee z$$



$$2. f(x,y,z) = (x \wedge y) \vee (y \wedge z)$$



$$3. f(x,y,z) = (x \wedge y) \oplus z$$



$$4. f(x,y,z) = (x \wedge y) \vee (x \wedge z) \vee (y \wedge z)$$

